Math 212

Homework Due Week 8 Monday

Winter 2026 **Prof. Chowdhury**

Note: The solutions were converted from my notes to LaTeX and partially generated using an AI tool. I have reviewed and edited them for accuracy.

Question 8001.

Use Gauss-Legendre quadrature to estimate $\int_{-1}^{1} x^2 e^x \, dx$ *in the following three ways:*

- *(a) use Gauss-Legendre quadrature with* $n = 8$ *(i.e., using roots of* φ_9*) on the entire interval* $[-1, 1]$*;*
- *(b) use Gauss-Legendre quadrature with* $n = 4$ *on each of the intervals* $[-1, 0]$ *and* $[0, 1]$ *and add the results;*
- *(c) use Gauss-Legendre quadrature with* $n = 2$ *on each of the intervals* $[-1, -1/2], [-1/2, 0], [0, 1/2]$ *and* $[1/2, 1]$ *and add the results.*

Compute the error in each case (you can find the exact value using integration by parts) and explain any trends you observe. Can you explain the trend using the theoretical error bound we proved in class?

Then, do the same for $\int_{-1}^{1} x^2 e^{|x|} \, dx$*. Did you find anything surprising? Explain what happened.*

Solution. We evaluate the performance of Gauss-Legendre quadrature for two different integrands. The results demonstrate the interplay between the degree of exactness and the regularity (smoothness) of the integrand.

(a) Smooth Integrand: $f(x) = x^2 e^x$

The exact value is $I_1 = e - 5e^{-1} \approx 0.87888462260183362738$.

Method	Absolute Error
(a) $n = 8$ (Roots of φ_9) on $[-1, 1]$	5.666×10^{-19}
(b) $n = 4$ (Roots of φ_5) on 2 sub-intervals	8.504×10^{-11}
(c) $n = 2$ (Roots of φ_3) on 4 sub-intervals	6.123×10^{-7}

Theoretical Explanation: According to the theorem proved in class, Gauss quadrature of order n has a **degree of exactness** of $2n + 1$. For $f \in C^{2n+2}[-1, 1]$, the error is:

$$I(f) - G_n(f) = \frac{f^{(2n+2)}(\xi)}{(2n+2)!} \int_{-1}^{1} \varphi_{n+1}^2(x) dx$$

In Method (a), the degree of exactness is 17. The high order of the derivative and the large (18!) denominator lead to near-machine precision. In Method (c), despite reducing the sub-interval width h (which usually helps in composite rules), the much lower degree of exactness ($2n+1=5$) results in a significantly larger error. For analytic functions, increasing the order n is generally more efficient than h-refinement.

(b) Non-smooth Integrand: $f(x) = x^2 e^{|x|}$

The exact value is $I_2 = 2(e-2) \approx 1.4365636569180904707$.

Method	Absolute Error
(a) $n=8$ (Roots of φ_9) on $[-1,1]$	2.070×10^{-4}
(b) $n=4$ (Roots of φ_5) on 2 sub-intervals	1.314×10^{-10}
(c) $n=2$ (Roots of φ_3) on 4 sub-intervals	9.815×10^{-7}

Theoretical Explanation: The error formula requires $f \in C^{2n+2}$. The function $f(x) = x^2 e^{|x|}$ is only in $C^2(-1,1)$, as $f'''(x)$ has a jump discontinuity at $x=0$.

- **Method (a)** fails to achieve its theoretical potential because the nodes span the singularity at $x=0$, violating the smoothness assumption required for degree 17 accuracy.
- **Methods (b) and (c)** partition the domain at $x=0$. On the resulting sub-intervals (e.g., $[0,1]$), the function is analytic. Thus, the error bound applies locally on each piece. Method (b) is vastly more accurate than (a) because it avoids integrating across the derivative discontinuity, allowing the $2n+1=9$ degree of exactness to be realized on each smooth segment.

■

Note: You can find the code in Canvas.

Question 8002.

Let

$$\mathrm{F} := \int_{-1}^{1} \frac{\mathrm{d}t}{\sqrt{(1-t^2)(1-t/2)}}$$

Use any numerical quadrature rule you like to compute this integral correctly to 10 *decimal places and* ***prove that your estimate is correct***.

Warning: *What happens to the integrand near $t = 1$? What can you do about this?*

Solution. The integrand has algebraic endpoint singularities of the form $(1-t^2)^{-1/2}$, which is precisely the Gauss–Chebyshev weight $w(t) = (1-t^2)^{-1/2}$. Writing

$$\mathrm{F} = \int_{-1}^{1} \frac{f(t)}{\sqrt{1-t^2}}\,dt, \qquad f(t) = \left(1 - \frac{t}{2}\right)^{-1/2},$$

the singularity is absorbed into the weight and f is analytic on $[-1, 1]$. The order-n Gauss–Chebyshev rule approximates F by

$$\mathrm{Q}_n = \frac{\pi}{n+1} \sum_{i=1}^{n+1} f(t_i), \qquad t_i = \cos\frac{(2i-1)\pi}{2(n+1)},$$

with all weights equal to $\pi/(n+1)$.

We derive a rigorous error bound. The standard error formula for an order-n Gaussian rule is

$$\mathrm{E}_n = \frac{f^{(2n+2)}(\xi)}{(2n+2)!} \int_{-1}^{1} \varphi_{n+1}(t)^2\, w(t)\,dt, \quad \xi \in (-1, 1),$$

where φ_{n+1} is the monic degree-$(n+1)$ orthogonal polynomial. For Chebyshev polynomials the monic form is $\varphi_{n+1} = 2^{-n}\mathrm{T}_{n+1}$, and

$$\int_{-1}^{1} \varphi_{n+1}(t)^2\, w(t)\,dt = 2^{-2n} \int_{-1}^{1} \mathrm{T}_{n+1}(t)^2\, w(t)\,dt = \frac{\pi}{2^{2n+1}},$$

so the error simplifies to

$$\mathrm{E}_n = \frac{f^{(2n+2)}(\xi)}{(2n+2)!} \cdot \frac{\pi}{2^{2n+1}}.$$

To bound $f^{(2n+2)}$ we need an explicit derivative formula. One can verify by induction that

$$f^{(n)}(t) = \frac{(2n)!}{8^n\, n!} \left(1 - \frac{t}{2}\right)^{-n-1/2},$$

and substituting $n \mapsto 2n+2$:

$$f^{(2n+2)}(t) = \frac{(4n+4)!}{64^{n+1}\,(2n+2)!}\left(1-\frac{t}{2}\right)^{-(4n+5)/2}.$$

Since $(1-t/2)^{-(4n+5)/2}$ is increasing on $[-1,1]$, the maximum occurs at $t=1$, where $(1-t/2)^{-(4n+5)/2} = 2^{2n+2}\sqrt{2}$, giving

$$\max_{t\in[-1,1]} f^{(2n+2)}(t) = \frac{(4n+4)!\,\sqrt{2}}{(2n+2)!\cdot 16^{n+1}}.$$

Substituting into the error formula:

$$|\mathrm{E}_n| \le \frac{1}{(2n+2)!}\cdot\frac{(4n+4)!\,\sqrt{2}}{(2n+2)!\cdot 16^{n+1}}\cdot\frac{\pi}{2^{2n+1}} = \binom{4n+4}{2n+2}\frac{\pi\sqrt{2}}{2^{6n+5}}.$$

Applying the standard bound $\binom{2m}{m} < 4^m$ with $m = 2n+2$:

$$|\mathrm{E}_n| < \frac{\pi\sqrt{2}}{2^{2n+1}}.$$

For ten decimal places we require $|\mathrm{E}_n| < \frac{1}{2}\times 10^{-10}$, i.e. $2^{2n+1} > 2\pi\sqrt{2}\times 10^{10} \approx 8.89\times 10^{10}$. Since $2^{37} \approx 1.37\times 10^{11}$, taking $2n+1 \ge 37$ suffices, i.e. $n \ge 18$.

The order-18 (19-point) Gauss–Chebyshev rule gives

$$\mathrm{F} \approx \frac{\pi}{19}\sum_{i=1}^{19}\left(1-\frac{1}{2}\cos\frac{(2i-1)\pi}{38}\right)^{-1/2} = 3.6638623767,$$

and the bound above guarantees $|\mathrm{E}_{18}| < 3.3\times 10^{-11}$, so this estimate is correct to 10 decimal places. ■

Question 8003.

Without worrying about whether it will work, try to use adaptive quadrature to estimate

$$\int_0^1 x\sin(1/x)\,\mathrm{d}x$$

to within an accuracy of 10^{-10}. *Explain (with theoretical justification) what happens and why.*

Solution. The integral $\int_0^1 x\sin(1/x)\,\mathrm{d}x$ is perfectly well-defined: since $|x\sin(1/x)| \le x$ on $(0,1]$ and $\lim_{x\to 0^+} x\sin(1/x) = 0$, the integrand extends to a continuous, bounded function on all of $[0,1]$. The difficulty is entirely one of *regularity* near $x = 0$.

Setting $f(x) = x\sin(1/x)$ for $x > 0$ and differentiating twice gives

$$f'(x) = \sin(1/x) - \frac{1}{x}\cos(1/x), \qquad f''(x) = -\frac{\sin(1/x)}{x^3},$$

and one can check by induction that $|f^{(n)}(x)| = \mathcal{O}(x^{1-2n})$ as $x \to 0^+$ for every $n \ge 1$. In particular, $|f^{(4)}(x)| = \mathcal{O}(x^{-7}) \to +\infty$.

Recall that you showed in the second lab that the error of Simpson's rule on an interval of width h satisfies

$$|\mathrm{E}_h| \approx \mathcal{O}(h^5)\left|f^{(4)}(\xi)\right|$$

for some ξ in the interval. But, on a subinterval $[\varepsilon, 2\varepsilon]$ near the origin, $|f^{(4)}(\xi)| = \mathrm{O}(\varepsilon^{-7})$, so

$$|\mathrm{E}_{[\varepsilon,2\varepsilon]}| = \mathrm{O}\left(\varepsilon^{-2}\right) \longrightarrow +\infty \quad \text{as } \varepsilon \to 0^+.$$

No matter how finely the adaptive algorithm bisects near the origin, the local error estimate *grows* rather than shrinks, so the tolerance criterion is never satisfied and the algorithm recurses indefinitely (or exhausts its subdivision budget without converging).

To compute the integral, one should handle the singularity analytically before applying quadrature. Choose a cutoff $\delta > 0$ and write

$$\int_0^1 f(x)\,\mathrm{d}x = \int_0^\delta f(x)\,\mathrm{d}x + \int_\delta^1 f(x)\,\mathrm{d}x.$$

The first piece satisfies $\left|\int_0^\delta f\right| \le \delta^2/2$, so choosing $\delta = \sqrt{2\cdot 10^{-10}} \approx 1.4\times 10^{-5}$ controls that term to within the target tolerance. On $[\delta, 1]$ all derivatives of f are bounded, and adaptive quadrature converges without difficulty. ■

Question 8004.

Show the initial value problem

$$y' = y - t\cos ty, \quad 0 \le t \le 2, \quad y(0) = 2$$

has a unique solution and is also well-posed.

Solution. Write the IVP in standard form $y' = f(t,y)$ with $f(t,y) = y(1 - t\cos t)$, on the infinite strip $D = [0,2] \times (-\infty, \infty)$. Since f is a product of continuous functions it is continuous on D, and

$$\frac{\partial f}{\partial y}(t,y) = 1 - t\cos t,$$

which is also continuous on D. By Picard's Theorem, the IVP has a unique C^1 solution on $[0,2]$.

It remains to show the problem is well-posed, for which we need a Lipschitz condition on f in y. For any $y_1, y_2 \in \mathbb{R}$ and $t \in [0,2]$,

$$|f(t,y_1) - f(t,y_2)| = |1 - t\cos t|\,|y_1 - y_2| \le \left(1 + 2\right)|y_1 - y_2| = 3|y_1 - y_2|,$$

where we used $|t\cos t| \le t \le 2$ on $[0,2]$. Thus f is Lipschitz in y on D with constant $L = 3$. By the well-posedness theorem (check notes), the IVP is well-posed.

■

Question 8005.

Let $h = 1/M$, where $M \geq 1$ is an integer, and let Euler's method be applied to calculate the estimates $\{y_n\}_{n=1,2,\ldots,M}$ of $y(nh)$ for each of the differential equations

$$y' = -\frac{y}{1+t} \quad \text{and} \quad y' = \frac{2y}{1+t}, \quad 0 \leq t \leq 1,$$

starting with $y_0 = y(0) = 1$ in both cases. By using induction and by cancelling as many terms as possible in the resultant products, deduce simple explicit expressions for $y_n, n = 1, 2, \ldots, M$, which should be free from summations and products of n terms. Hence deduce the exact solutions of the equations from the limit $h \to 0$. Verify that the magnitude of the errors $y_n - y(nh), n = 1, 2, \ldots, M$, is at most $\mathcal{O}(h)$.

Solution. $\boxed{\text{First ODE}}$

Consider the differential equation $y' = -y/(1+t)$ for $0 \leq t \leq 1$, subject to $y(0) = 1$. The Euler method iteration, defined by $y_{n+1} = y_n + hf(t_n, y_n)$ with $t_n = nh$, yields:

$$y_{n+1} = y_n - h\frac{y_n}{1+nh} = y_n\left(1 - \frac{h}{1+nh}\right) = y_n\left(\frac{1+(n-1)h}{1+nh}\right).$$

By induction, it immediately follows that the explicit sequence is given by

$$y_n = \frac{1-h}{1+(n-1)h}, \quad \text{for } n = 0, 1, \ldots, M.$$

To deduce the exact solution via the limit, let $h \to 0$ such that $nh = t$ remains fixed. We obtain:

$$\lim_{\substack{h\to 0\\ nh\to t}} y_n = \lim_{h\to 0} \frac{1-h}{1+t-h} = \frac{1}{1+t}.$$

We can check that $y(t) = \dfrac{1}{1+t}$ does in fact satisfy the ODE.

Therefore, the numerical error at step n is:

$$e_n = y_n - y(nh) = \frac{1-h}{1+(n-1)h} - \frac{1}{1+nh} = \frac{(1-h)(1+nh) - (1+nh-h)}{(1+(n-1)h)(1+nh)}.$$

Thus,

$$|e_n| = \frac{nh^2}{(1+(n-1)h)(1+nh)}.$$

Since $n \geq 1$, we have $(1+(n-1)h) \geq 1$ and $(1+nh) \geq 1$. Furthermore, $nh \leq 1$ because $t \in [0,1]$. This implies $|e_n| \leq nh^2 \leq h$, confirming that $|e_n| = \mathcal{O}(h)$.

$\boxed{\text{Second ODE}}$

Consider the differential equation $y' = 2y/(1+t)$ for $0 \leq t \leq 1$, with $y(0) = 1$. The associated Euler iteration is:

$$y_{n+1} = y_n + h\frac{2y_n}{1+nh} = y_n\left(1 + \frac{2h}{1+nh}\right) = y_n\left(\frac{1+(n+2)h}{1+nh}\right).$$

Expressing y_n as a product of consecutive updates yields a telescoping product:

$$y_n = y_0 \prod_{k=0}^{n-1} \frac{1+(k+2)h}{1+kh} = \frac{(1+nh)(1+(n+1)h)}{1+h} = \frac{1+(2n+1)h+n(n+1)h^2}{1+h}.$$

Evaluating the continuum limit $h \to 0$ subject to $nh = t$:

$$\lim_{\substack{h\to 0\\ nh\to t}} y_n = \lim_{h\to 0} \frac{(1+t)(1+t+h)}{1+h} = (1+t)^2.$$

The function $y(t) = (1+t)^2$ does satisfy the IVP.

The error $e_n = y_n - y(nh)$ evaluates to:

$$\begin{aligned}
e_n &= \frac{(1+nh)(1+nh+h)}{1+h} - (1+nh)^2 \\
&= (1+nh)\left[\frac{1+nh+h-(1+nh)(1+h)}{1+h}\right] \\
&= (1+nh)\left[\frac{1+nh+h-(1+h+nh+nh^2)}{1+h}\right] \\
&= \frac{-nh^2(1+nh)}{1+h}.
\end{aligned}$$

Assessing the magnitude, taking into account $nh \le 1$ and $(1+h)^{-1} \le 1$, we find:

$$|e_n| = \frac{nh^2(1+nh)}{1+h} \le h \cdot (1+1) \cdot 1 = 2h.$$

This formally guarantees the bound $|e_n| = \mathcal{O}(h)$.

■